THE UNIVERSITY OF SYDNEY

ELEC5211: Power System Dynamics and Control

PROJECT REPORT

GROUP 16

Multi-Machine Power System Steady-State and Dynamic Stability Analysis

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1. Introduction

This project investigates the steady-state and dynamic behaviour of the IEEE 3-machine 9-bus power system. The main objectives are to perform steady-state power flow analysis, study the effects of voltage references, loading level and reactive compensation, initialize the multi-machine dynamic model, simulate three-phase fault responses, evaluate post-fault stability with and without damping, and estimate the critical clearing time (CCT).

The steady-state power flow studies are carried out in MATLAB using MATPOWER. The dynamic simulations are implemented using a classical second-order generator model and solved by time-domain numerical integration. To make the report self-contained and easy to follow, the report is organized according to the project tasks. Each task includes the modelling method, simulation results, and discussion of the observed behaviour.

This report also pays special attention to unit consistency and the formulation of the dynamic model. All dynamic quantities are expressed in the per-unit system on the selected system base, and the differential equations are arranged into a state-space form suitable for numerical simulation.

2. System Description and Modelling Assumptions

2.1 System Overview

The study is based on the IEEE 3-machine 9-bus power system, which is a standard benchmark system for power flow and transient stability analysis. The system operates at 50 Hz and contains three synchronous generators connected at buses 1, 2 and 3. These generators are connected to the transmission network through step-up transformers. The transmission network supplies three aggregated loads located at buses 5, 7 and 9 in the implemented MATPOWER model.

In this project, the system is studied under both steady-state and dynamic conditions. The steady-state part uses power flow calculations to evaluate bus voltages, generator outputs and branch power flows. The dynamic part uses a classical second-order generator model to study rotor-angle and rotor-speed responses during and after a three-phase fault.

2.2 Key Operating Parameters

Base case operating conditions are specified as follows:

Parameter	Value
System base power (S_B)	100 MVA
System frequency (f_0)	50 Hz
Load at Bus 5	90+j30 MVA
Load at Bus 7	100+j30 MVA
Load at Bus 9	125+j50 MVA
Generator 1 voltage set-point	$V_{G1} = 1.04 p.u.$
Generator 2 voltage set-point	$V_{G2} = 1.025 p.u.$
Generator 3 voltage set-point	$V_{G3} = 1.025 p.u.$

The total base-case load is therefore

$$S_{load,total} = 315 + j110\ MVA$$

which is used as the reference operating condition for the power flow and dynamic initialization studies.

2.3 Units and Base Values

To avoid unit inconsistency in the dynamic model, all quantities used in the differential equations are expressed in the per-unit system on the 100 MVA system base. The power flow results are initially obtained in MW and MVar, and then converted into per-unit values before being used in the swing equations. Rotor angles are solved in radians in MATLAB and converted into degrees only for plotting and discussion.

The main units used in the simulation are summarized in Table 1

Table 1. Units and base values used in the dynamic simulation.

Quantity	Unit used in simulation
S_B	100 MVA
P_G, Q_G, P_m, P_e	p.u. on 100 MVA base in dynamic model
V, E	p.u.
x_d'	p.u. on 100 MVA base
δ	rad in equations and MATLAB; degree in figures
ω	p.u.
ω_0	rad/s
H	s
$T_J = 2H$	s
t	s

2.4 Modelling Assumptions

The following assumptions are used in this project: For steady-state power flow

analysis, the Newton-Raphson power flow method is used. Bus 1 is selected as the slack bus, buses 2 and 3 are treated as PV buses, and the remaining buses are treated as PQ buses.

The generator dynamic model is represented by the classical second-order model. The transient internal voltage magnitude E' is assumed to remain constant during the electromechanical transient simulation.Loads are represented as constant impedance loads in the dynamic simulation. This allows the network admittance matrix to be updated for pre-fault, fault-on and post-fault conditions.

The three-phase fault is represented by applying a large shunt admittance at Bus 7. After fault clearing, the fault is removed and the line connected to Bus 7 in the implemented network topology is opened to represent the post-fault condition.The damping coefficient is set to $D = 0$ for the undamped case and $D = 10$ for the damped case in the post-fault stability comparison.

3. Program Structure and Numerical Method

3.1 Overall Program Structure

The simulation program is organized into five main modules. This modular structure makes it easier to test each part of the project separately and ensures that the results for each task can be clearly identified.

Power flow calculation module

Sensitivity analysis module

Dynamic initialization module

Fault-on and post-fault simulation module

Critical clearing time search module

The power flow module provides the steady-state operating point. The sensitivity analysis module studies how voltage set-points, reference angle, loading level and reactive compensation affect the system. The dynamic initialization module converts the power flow result into the initial conditions of the generator dynamic model. The fault simulation module solves the multi-machine swing equations under fault-on and post-fault network conditions. Finally, the CCT search module estimates the maximum allowable fault clearing time for transient stability.

3.2 Key Module Descriptions

3.2.1 Power Flow Block

Loads the standard IEEE 9-bus system case

Modifies load parameters and generator voltage setpoints according to test scenarios

Runs the Newton-Raphson power flow solver

Extracts and stores bus voltages, generator outputs, and branch power flows

3.2.2 Sensitivity Analysis Block

Automates parameter variation for different test scenarios

Performs batch simulations for generator voltage setpoint changes, reference angle adjustments, load scaling, and reactive compensation

Generates comparative plots and sensitivity curves

3.2.3 Dynamic Initialization Block

Converts steady-state power flow results to dynamic model initial conditions

Calculates generator currents, transient internal voltages, and rotor angles

Sets initial mechanical power equal to electrical power to ensure equilibrium

Verifies that all state derivatives are zero at the initial operating point

3.2.4 Fault Simulation Module

The fault simulation module constructs separate reduced network admittance matrices for the fault-on and post-fault conditions. The multi-machine swing equations are solved during the fault-on period first. Then, the state at the fault clearing instant is used as the initial condition for the post-fault simulation. The module records generator rotor angles, rotor speeds and bus voltage magnitudes for stability analysis.

3.2.5 CCT Search Block

Iteratively tests different fault clearing times

Judges system stability based on relative rotor angle deviations

Uses a binary search algorithm to refine the CCT estimate with high precision

3.3 Dynamic Model and Governing Equations

The dynamic simulation in this project is based on the classical second-order generator model. In this model, each synchronous generator is represented by a constant transient internal voltage behind the transient reactance \(x'_d\). The main dynamic variables are the rotor angle and rotor speed of each generator. Therefore, for the 3-machine system, the state vector is defined as

$$x = [\delta_1\ \delta_2\ \delta_3\ \omega_1\ \omega_2\ \omega_3]^{\mathrm{T}}$$

where δ_i is the rotor angle of generator i, and ω_i is the rotor speed in per unit.

The base-case power flow result is used to initialize the dynamic model. For generator i, the complex power output is

$$S_i = P_{G,i} + jQ_{G,i}$$

where $P_{G,i}$ and $Q_{G,i}$ are converted into per-unit values on the 100 MVA system base before being used in the dynamic model. From the complex power relationship

$$S_i = V_i I_i^*$$

the generator current is calculated as

$$I_i = \left(\frac{S_i}{V_i}\right)^*$$

The transient internal voltage of generator \(i\) is then calculated by

$$E'_i = V_i + jx'_{d,i} I_i$$

and the initial rotor angle is obtained from the phase angle of the internal voltage:

$$\delta_i = \angle E'_i$$

The inertia constant is calculated from the stored energy data using

$$H_i = \frac{W_{k,i}}{S_B}$$

where$W_{k,i}$ is the stored energy of generator i at rated speed and $S_B = 100\ MVA$ is the system base power. The inertia time constant used in the swing equation is

$$T_{J,i} = 2H_i$$

The rotor-angle equation is

$$\frac{d\delta_i}{dt} = \omega_0(\omega_i - 1)$$

where $\omega_0 = 2\pi f_0$ is the synchronous angular frequency and $f_0 = 50\ Hz$. The rotor-speed equation is

$$T_{J,i}\frac{d\omega_i}{dt} = P_{m,i} - P_{e,i} - D_i(\omega_i - 1)$$

where $P_{m,i}$ is the mechanical input power,$P_{e,i}$ is the electrical output power, and D_i is the damping coefficient.

To make the differential equations solvable by an ordinary differential equation solver, the electrical output power$P_{e,i}$ must be calculated from the state variables. For a given rotor-angle vector, the internal voltage vector is

$$E = [E'_1 e^{j\delta_1}\ E'_2 e^{j\delta_2}\ E'_3 e^{j\delta_3}]^{\mathrm{T}}$$

After Kron reduction of the network admittance matrix, the generator internal current vector is calculated by

$$I = Y_{red} E$$

where Y_{red} is the reduced admittance matrix seen from the generator internal buses. Therefore, the electrical output power of generator i is

$$P_{e,i} = Re(E_i I_i^*)$$

Since $P_{e,i}$ is calculated from the rotor-angle states through Y_{red}, the final dynamic model can be expressed in the standard state-space form

$$\dot{x} = f(x)$$

This form is suitable for time-domain numerical simulation because once the initial state $x(0)$ is known, the derivative of every state variable can be calculated directly.

3.4 Numerical Solution Method

The steady-state power flow calculation is solved using the Newton-Raphson method implemented in MATPOWER. The power flow solution provides the base-case bus voltage magnitudes and angles, generator active and reactive power outputs, and branch active and reactive power flows. These values are then used as the reference operating point for the sensitivity analysis and dynamic initialization.

The time-domain dynamic simulations are solved using MATLAB ode45. The

fault-on and post-fault simulations are solved in two stages. First, the swing equations are integrated from the fault occurrence time to the fault clearing time using the fault-on reduced network admittance matrix. Then, the final state at the clearing instant is used as the initial condition for the post-fault simulation with the post-fault network admittance matrix.

For the CCT search, different fault clearing times are tested under the undamped condition. The system is considered unstable if any relative rotor angle with respect to Generator 1 exceeds 180 degrees during the simulation window.

4. Steady-State Power Flow Analysis （Task 1）

This section presents the steady-state power flow analysis of the IEEE 3-machine 9-bus system. The objectives of Task 1 are to obtain the base-case power flow solution, examine the effects of generator voltage set-points, and investigate the influence of the reference bus angle. The main quantities observed include bus voltage magnitudes and angles, generator active and reactive power outputs, and branch active and reactive power flows.

4.1 Base Case Power Flow

The base-case power flow calculation converged successfully. The total system load is

$$S_{load,total} = 315 + j110\ MVA$$

The bus voltage profile is shown in Figure 1. All bus voltage magnitudes remain within the normal operating range of 0.95 p.u. to 1.05 p.u. The lowest voltage occurs at Bus 9, with a voltage magnitude of approximately 0.9965 p.u. Therefore, Bus 9 is identified as the weakest voltage bus in the base-case operating condition.

The generator output results were also checked. Generator 1 acts as the slack generator and balances the system active power mismatch and network losses, while Generators 2 and 3 operate as PV generators with specified voltage magnitudes. The branch active and reactive power flows were used to confirm the direction and magnitude of power transfer in the network. These base-case bus, generator and branch results provide the reference for the following sensitivity and dynamic studies.

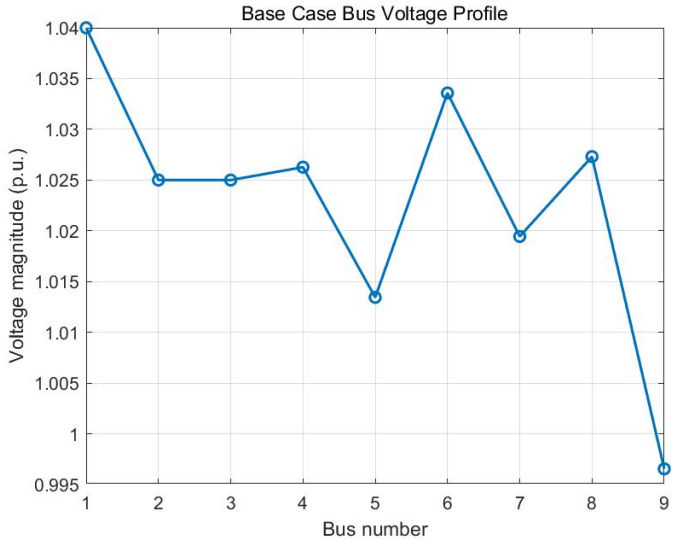


Figure 1. Base-case bus voltage profile of the IEEE 3-machine 9-bus system.

Table 2. Base-case generator active and reactive power outputs.

Generator bus	$(P_G)p.u.$	$(Q_G)p.u.$
1	0.7162	0.2617
2	1.6300	0.0412
3	0.8500	-0.1300

4.2 Effect of Generator Voltage Set-Points

To evaluate the effect of generator voltage regulation on the system voltage profile, three voltage set-point cases are considered:

$$V_{G,base} = [1.04\ 1.025\ 1.025]$$

$$V_{G,high} = [1.06\ 1.04\ 1.04]$$

$$V_{G,low} = [1.02\ 1.01\ 1.01]$$

As shown in Figure 2, increasing the generator voltage set-points raises the voltage magnitudes across the network, while decreasing the set-points lowers the voltage profile. This is because the generator buses are treated as PV buses, where the voltage magnitude is controlled and the reactive power output adjusts to support the specified voltage.

Figure 3 shows that changing the generator voltage set-points also changes the

generator reactive power outputs. The active power outputs remain nearly unchanged because the active power dispatch is not directly changed in this test. Therefore, this test shows that generator voltage references mainly affect the voltage profile and reactive power distribution.

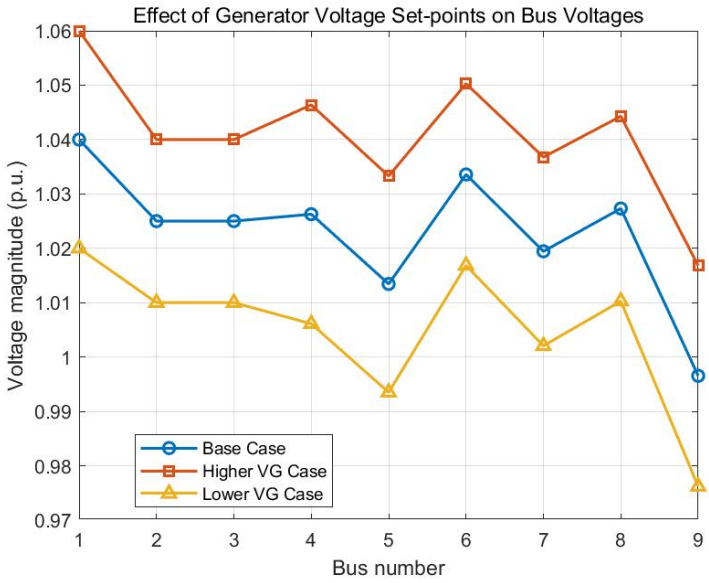


Figure 2. Effect of generator voltage set-points on bus voltage magnitudes.

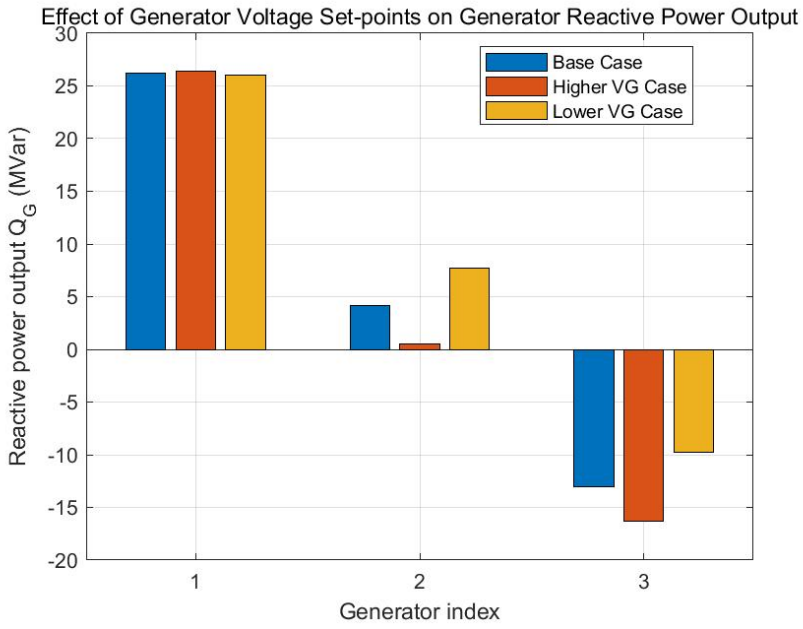


Figure 3. Effect of generator voltage set-points on generator reactive power outputs.

4.3 Effect of Reference Bus Angle

The reference angle of Bus 1 is adjusted to three values:

$$\theta_{ref} = 0°,\ +10°,\ -10°$$

The results are shown in Figures 4 and 5. Changing the reference bus angle shifts the voltage angles of all buses by approximately the same amount. However, the voltage magnitudes remain almost unchanged. The generator outputs and branch power flows are also essentially unchanged.

This is because the reference bus angle only defines the angular reference of the phasor coordinate system. It does not represent a physical change in the network operating condition. Therefore, changing the reference angle affects the numerical values of voltage angles, but it does not change the actual power flow solution.

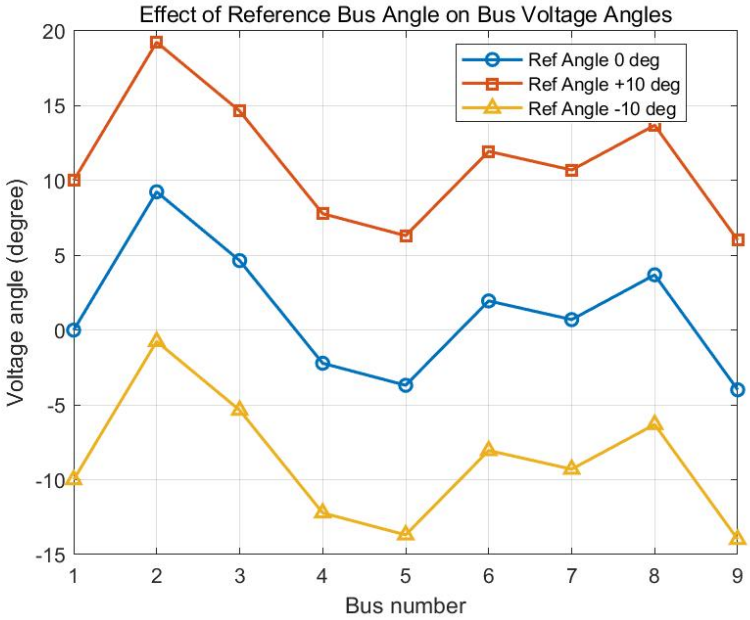


Figure 4. Effect of reference bus angle on bus voltage angles.

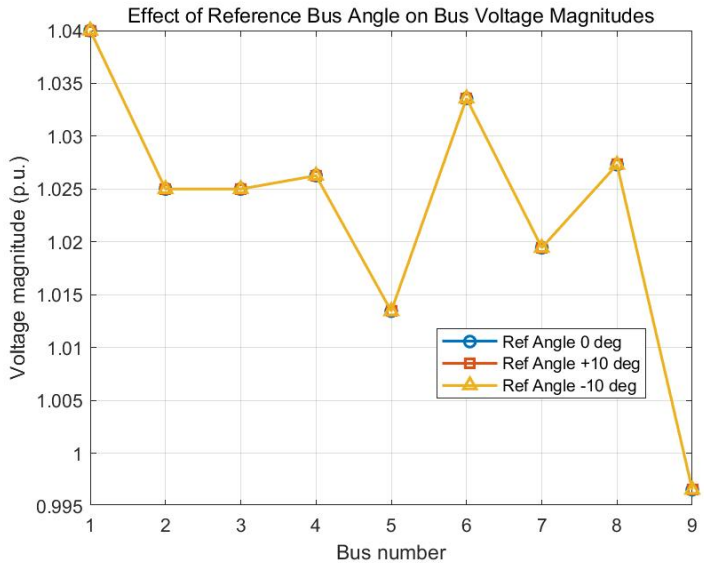


Figure 5. Effect of reference bus angle on bus voltage magnitudes.

Therefore, Task 1 shows that generator voltage set-points mainly affect bus voltage magnitudes and generator reactive power outputs, while the reference bus angle only changes the angular reference and does not change the physical power flow condition.

5. Effect of System Loading on Voltage Level （Task 2）

This section investigates the influence of system loading on the voltage level of the network. The objective is to identify how bus voltage magnitudes change when the active and reactive loads are increased or decreased from the base-case operating condition.

5.1 Scenario Design

In this task, both active and reactive loads are scaled by the same loading factor λ. Four loading levels are considered:

$$\lambda = 0.8,\ 1.0,\ 1.2,\ 1.4$$

The corresponding loading cases are 80%, 100%, 120%, and 140% of the base-case load. The load scaling equations are

$$P_{D,i}^{new} = \lambda P_{D,i}^{base}$$

$$Q_{D,i}^{new} = \lambda Q_{D,i}^{base}$$

where $P_{D,i}^{base}$ and $Q_{D,i}^{base}$ are the base-case active and reactive loads at bus i. During this study, the generator voltage set-points are kept unchanged:

$$V_{G1} = 1.04, \qquad V_{G2} = 1.025, \qquad V_{G3} = 1.025$$

This allows the effect of loading level on bus voltage magnitudes to be observed independently.

5.2 Simulation Results

Figure 6 shows the bus voltage magnitudes under different loading levels. As the load level increases from 80% to 140%, the voltage magnitudes decrease across the network. The voltage reduction is more significant at the load buses and electrically weaker buses.

Bus 9 has the lowest voltage magnitude under the heavy loading cases. In the 140% loading case, the voltage at Bus 9 drops to approximately 0.964 p.u., which makes Bus 9 the weakest voltage bus in this study.

Figure 7 shows the minimum bus voltage under each loading level. The minimum bus voltage decreases continuously as the loading level increases. This confirms that heavier system loading weakens the voltage profile of the network.

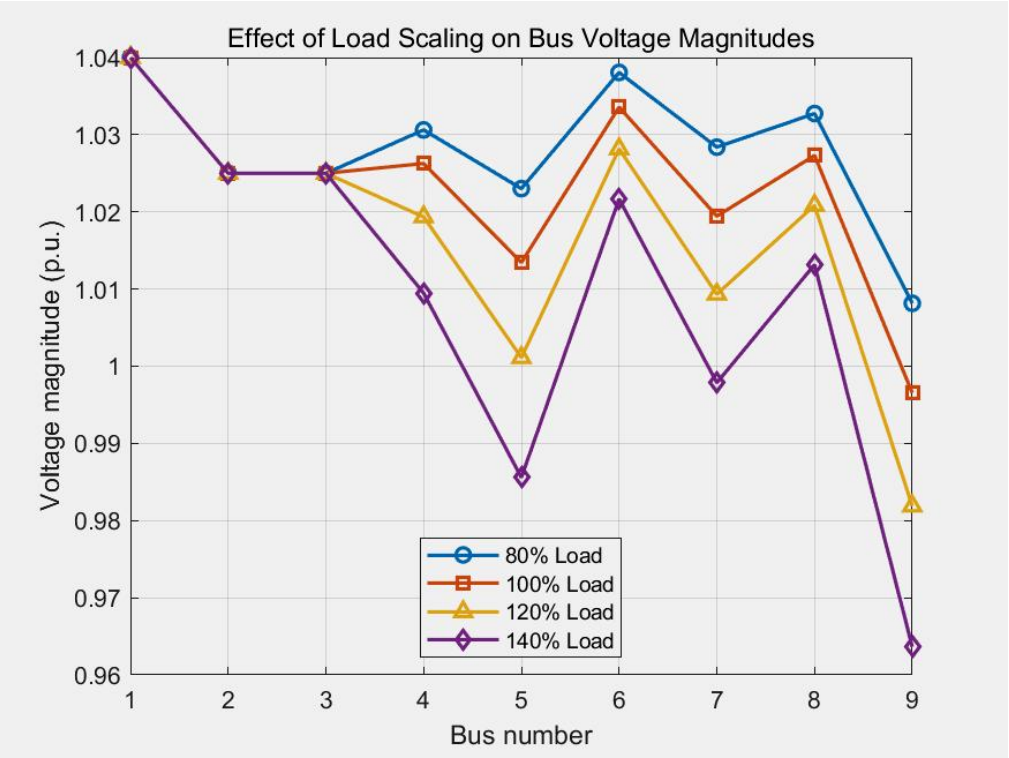


Figure 6. Effect of load scaling on bus voltage magnitudes.

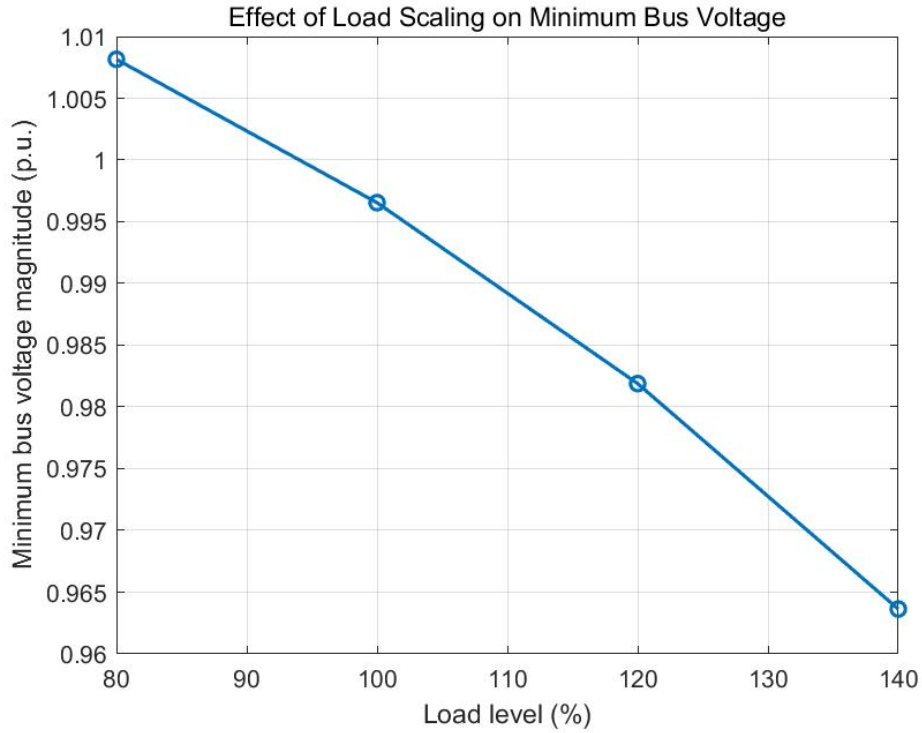


Figure 7. Effect of load scaling on minimum bus voltage.

5.3 Discussion

The simulation results show that system loading has a clear influence on the network voltage level. When the active and reactive loads increase, the currents flowing through the transmission lines and transformers also increase. As a result, the voltage drops across the network impedances become larger, causing lower bus voltage magnitudes.

The voltage drop is most obvious at Bus 9. This is because Bus 9 is a load bus and is electrically weaker than the generator buses. Under heavy loading, the local reactive power demand and network voltage drop become more severe, so Bus 9 becomes the limiting bus in the voltage profile.

These results indicate that excessive loading can reduce the voltage security margin of the system. If the loading level continues to increase, additional voltage support may be required. Possible measures include generator reactive power support, shunt capacitor compensation, or other voltage control devices.

Therefore, Task 2 confirms that heavier system loading weakens the voltage profile, especially at load buses such as Bus 9.

6. Reactive Power Compensation Analysis (Task 3)

This section investigates the effect of shunt reactive power compensation on the system voltage profile. Based on the loading analysis in Task 2, Bus 9 is selected as the compensation location because it has the lowest voltage magnitude under heavy loading conditions.

6.1 Scenario Design

Four reactive compensation cases are tested at Bus 9:

$$Q_{comp} = -50,\ 0,\ +50,\ +100\ MVar$$

In this report, negative compensation represents an inductive reactor that absorbs reactive power, while positive compensation represents a shunt capacitor that injects reactive power. The case $Q_{comp} = 0$ is the base case without additional compensation.

All other operating conditions, including load level and generator voltage set-points, are kept the same as the base case. Therefore, the voltage changes observed in this task are mainly caused by the reactive compensation at Bus 9.

6.2 Simulation Results

Figure 8 shows the bus voltage magnitudes under different reactive compensation cases. When a 50 MVar reactor is connected at Bus 9, the voltage profile decreases. This is because the reactor absorbs reactive power from the system and increases the reactive power burden of the network.

When capacitive compensation is applied, the bus voltage magnitudes increase. The 50 MVar capacitor improves the voltage profile effectively by injecting reactive power locally at Bus 9. The 100 MVar capacitor further increases the voltage magnitude at Bus 9, but it also pushes the local voltage close to 1.10 p.u., which indicates a possible overvoltage problem.

Figure 9 shows the minimum bus voltage under different compensation levels. The minimum voltage increases as the compensation changes from reactive power absorption to reactive power injection. This confirms that capacitive compensation can improve the low-voltage condition of the system.

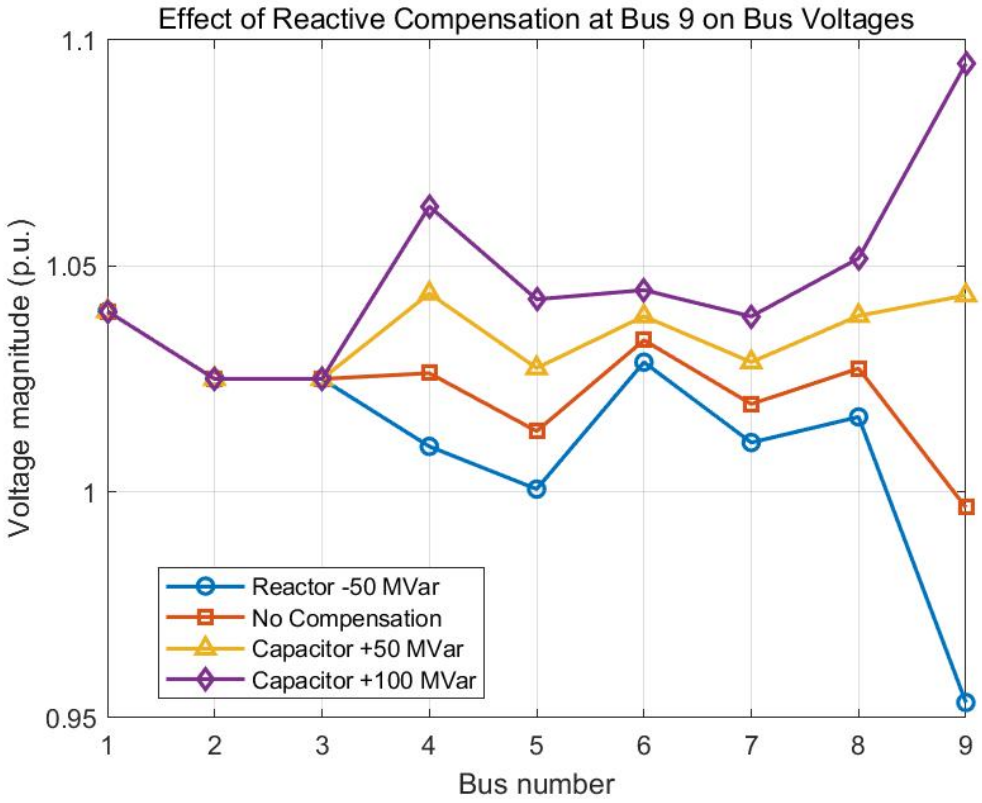


Figure 8. Effect of reactive compensation at Bus 9 on bus voltage magnitudes.

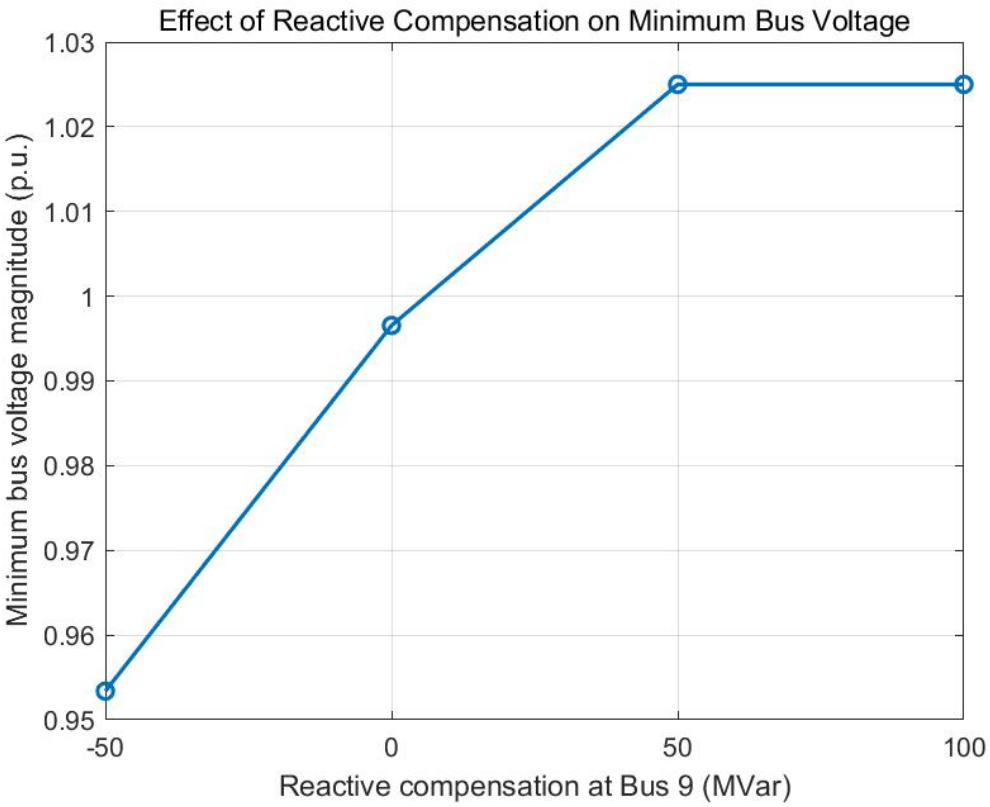


Figure 9. Effect of reactive compensation on minimum bus voltage.

6.3 Discussion

The simulation results show that reactive power compensation has a significant influence on the voltage profile. The reactor case reduces the voltage magnitude because it absorbs reactive power. In contrast, the capacitor cases raise the voltage profile because they inject reactive power close to the weak bus.

The improvement is most obvious around Bus 9, where the compensation device is connected. This demonstrates the local nature of reactive power support: placing compensation near a weak bus is more effective for improving the voltage at that location.

However, the 100 MVar capacitor case also shows that excessive capacitive compensation may cause local overvoltage. Therefore, reactive compensation should not be increased without limitation. A suitable compensation size should improve the voltage profile while keeping all bus voltages within an acceptable operating range.

Therefore, Task 3 shows that capacitive compensation can improve low-voltage conditions, while excessive compensation may create local overvoltage.

7. Dynamic Model Initialization and Equilibrium Check (Task 4)

This section initializes the classical dynamic model from the base-case power flow result. The objective of this task is to convert the steady-state power flow solution into the initial dynamic states of the three generators and then verify whether the initial operating point is an equilibrium of the swing-equation model.

In power system dynamic analysis, the power flow solution provides the steady-state operating point. Therefore, the bus voltages, generator active power outputs and generator reactive power outputs obtained from the base-case power flow are used to calculate the generator internal transient voltages, rotor angles and mechanical input powers.

7.1 Initialization Methodology

For each generator, the complex power output is first obtained from the base-case power flow result:

$$S_i = P_{G,i} + jQ_{G,i}$$

where $P_{G,i}$ and $Q_{G,i}$ are converted to per-unit values on the 100 MVA system base. Since the complex power satisfies

$$S_i = V_i I_i^*$$

the generator stator current is calculated as

$$I_i = \left(\frac{S_i}{V_i}\right)^*$$

The transient internal voltage of generator i is calculated using the classical generator model:

$$E'_i = V_i + jx'_{d,i} I_i$$

The initial rotor angle is then obtained from the angle of the internal transient

voltage:

$$\delta_i = \angle E'_i$$

The generator transient reactances x'_d and stored energy values are taken from the generator data. Since the system base is 100 MVA, the inertia constant on the system base is calculated as

$$H_i = \frac{W_{k,i}}{S_B}$$

where $W_{k,i}$ is the stored energy at rated speed and $S_B = 100\ MVA$. The inertia time constant used in the swing equation is

$$T_{J,i} = 2H_i$$

At the initial steady-state operating point, the rotor speed is set to synchronous speed:

$$\omega_i = 1$$

The mechanical input power is initialized to be equal to the electrical output power:

$$P_{m,i} = P_{e,i}$$

Therefore, the equilibrium condition of the swing equation is checked by

$$P_{m,i} - P_{e,i} - D_i(\omega_i - 1) = 0$$

If this mismatch is zero or sufficiently close to zero for all generators, the initial state is a valid equilibrium point for the dynamic model.

7.2 Initialization Results

The calculated initial conditions are summarized in Table 3. The results show the generator active and reactive power outputs, internal transient voltage magnitudes, initial rotor angles, inertia constants and equilibrium mismatches.

Table 3. Dynamic initialization and equilibrium check.

Gen	$P_G(p.u.)$	$Q_G(p.u.)$	$E'(p.u.)$	$\delta(deg)$	$H(s)$	$P_m - P_e(p.u.)$
1	0.7162	0.2617	1.0561	2.272	23.64	0
2	1.6300	0.0412	1.0473	19.728	6.40	0
3	0.8500	-0.1300	1.0132	13.181	3.01	0

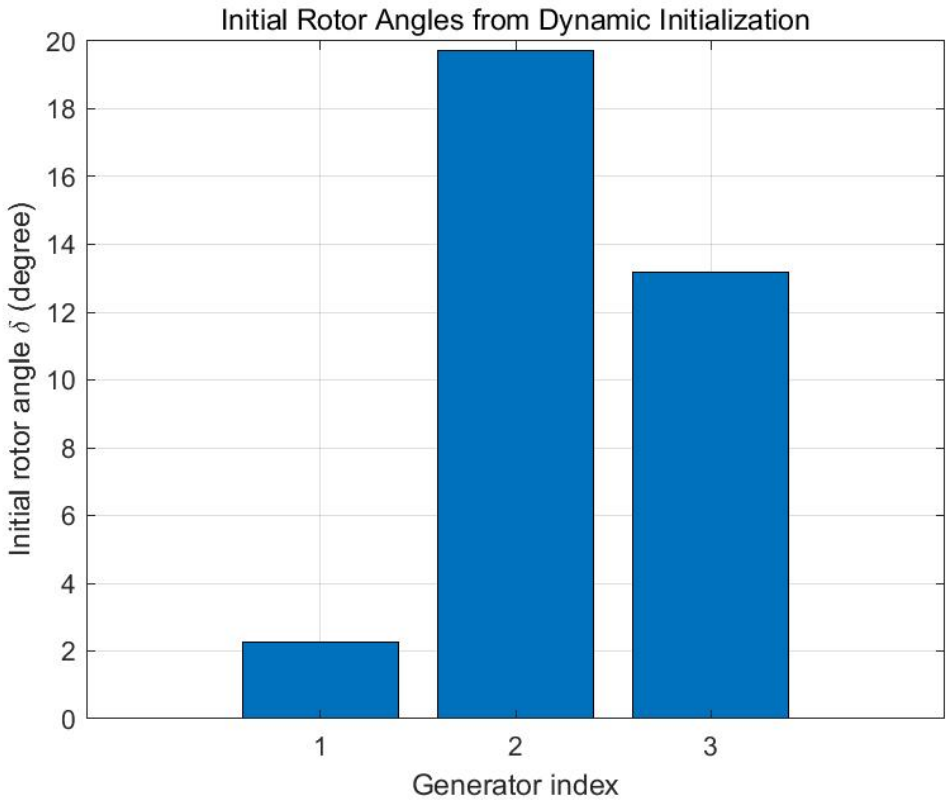


Figure 10. Initial rotor angles of the three generators.

7.3 Equilibrium Verification

At the initial operating point, the rotor speeds are set to $\omega_i = 1$ p.u. and the mechanical input powers are set equal to the electrical output powers. Therefore, the right-hand side of the swing equation should be zero:

$$P_{m,i} - P_{e,i} - D_i(\omega_i - 1) = 0$$

Since $\omega_i = 1$, the damping term is zero at the initial operating point. The equilibrium check is therefore mainly determined by the mismatch between $P_{m,i}$ and $P_{e,i}$.

As shown in Table 3, $P_m - P_e = 0$ for all three generators. This confirms that the base-case power flow solution has been correctly converted into a valid equilibrium point for the dynamic model. The initialized model can therefore be used as the starting point for the following fault-on and post-fault dynamic simulations.

Therefore, Task 4 confirms that the base-case power flow solution is a valid equilibrium point for the multi-machine dynamic model.

8. Three-Phase Fault-On Dynamic Simulation (Task 5)

This section simulates the system dynamic response during a three-phase fault. The purpose of this task is to observe how the generator rotor angles, generator rotor speeds, and bus voltage magnitudes change during the disturbance period.

8.1 Fault Modelling

In this simulation, the three-phase fault is represented by applying a large shunt admittance at Bus 7. This approximates a solid three-phase short circuit at the fault location. The fault admittance used in the simulation is

$$Y_{fault} = 10^6\ p.u.$$

The fault starts at t=0 s. Since the system frequency is 50 Hz and the fault is cleared after 4 cycles in the following task, the fault-on duration used here is

$$t_{fault} = \frac{4}{50} = 0.08\ s$$

During the fault-on period, the network admittance matrix is changed to represent the short-circuit condition. The generator mechanical input power is assumed to remain constant during this short electromechanical transient. The generator dynamics are governed by the swing equations:

$$\frac{d\delta_i}{dt} = \omega_0(\omega_i - 1)$$

$$T_{J,i}\frac{d\omega_i}{dt} = P_{m,i} - P_{e,i} - D_i(\omega_i - 1)$$

where $P_{e,i}$ is recalculated from the fault-on reduced network admittance matrix.

8.2 Dynamic Response Characteristics

The fault-on dynamic responses are shown in Figures 11 to 13. Figure 11 shows the generator rotor angles during the fault. Figure 12 shows the generator rotor speeds during the same period. Figure 13 shows the bus voltage magnitudes during the fault.

8.2.1 Generator Rotor Angles

Figure 11 shows that the rotor angles increase during the fault-on period. This happens because the three-phase fault reduces the electrical power transfer capability of the network. As a result, the electrical output power $P_{e,i}$ of the generators decreases while the mechanical input power $P_{m,i}$ remains approximately constant. The difference between $P_{m,i}$ and $P_{e,i}$ produces positive accelerating power.

The rotor-angle increase is not the same for all generators. Generator 1 has the largest inertia constant, so its rotor angle changes more slowly. Generators 2 and 3 have smaller inertia constants and therefore show larger angle changes during the fault period.

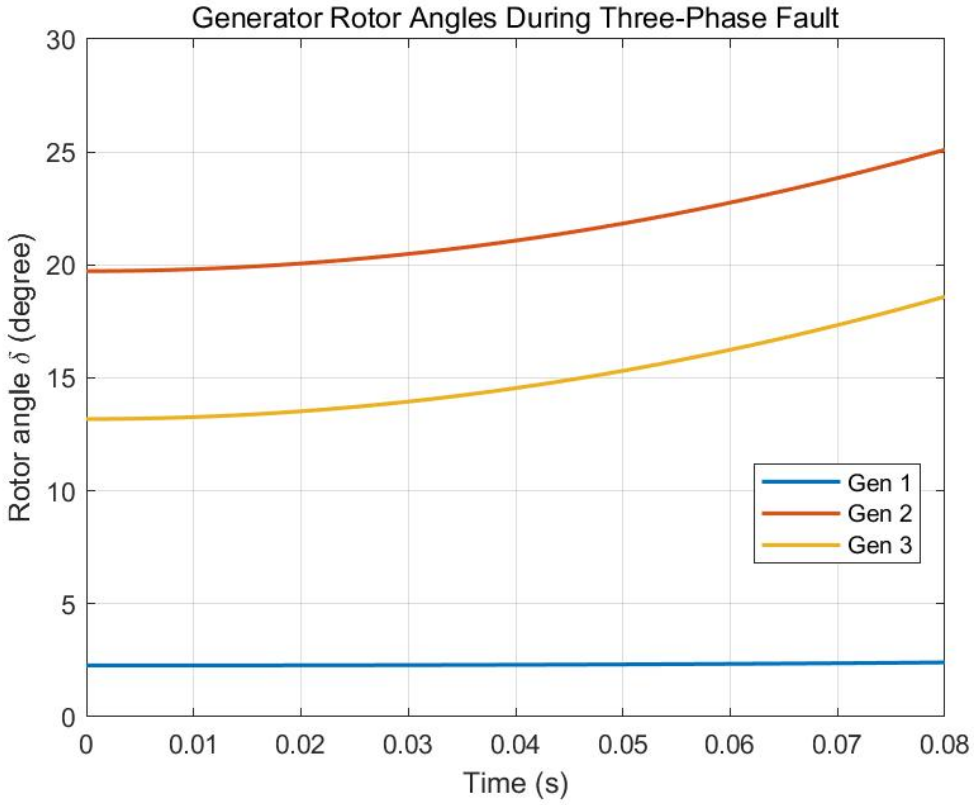


Figure 11. Generator rotor angles during the three-phase fault.

8.2.2 Generator Rotor Speeds

Figure 12 shows that the generator rotor speeds deviate from the synchronous speed during the fault. The rotor speed of Generator 1 changes only slightly, while the speeds of Generators 2 and 3 increase more clearly. This is consistent with the rotor-angle response and the different inertia constants of the generators.

The speed increase indicates that the generators are accelerating during the fault-on period. Since the fault duration is short, the response mainly shows the initial acceleration caused by the fault rather than a complete post-fault oscillation.



Figure 12. Generator rotor speeds during the three-phase fault.

8.2.3 Bus Voltage Magnitudes

Figure 13 shows that the voltage at the faulted bus collapses close to zero during the three-phase fault. This is the expected behaviour for a solid three-phase short circuit because the fault creates a very low-impedance path to ground.

The voltages at other buses are also depressed during the fault. The buses electrically closer to the fault location experience more severe voltage reduction, while the generator terminal buses are less severely affected because they are closer to the internal generator voltage sources in the classical model.

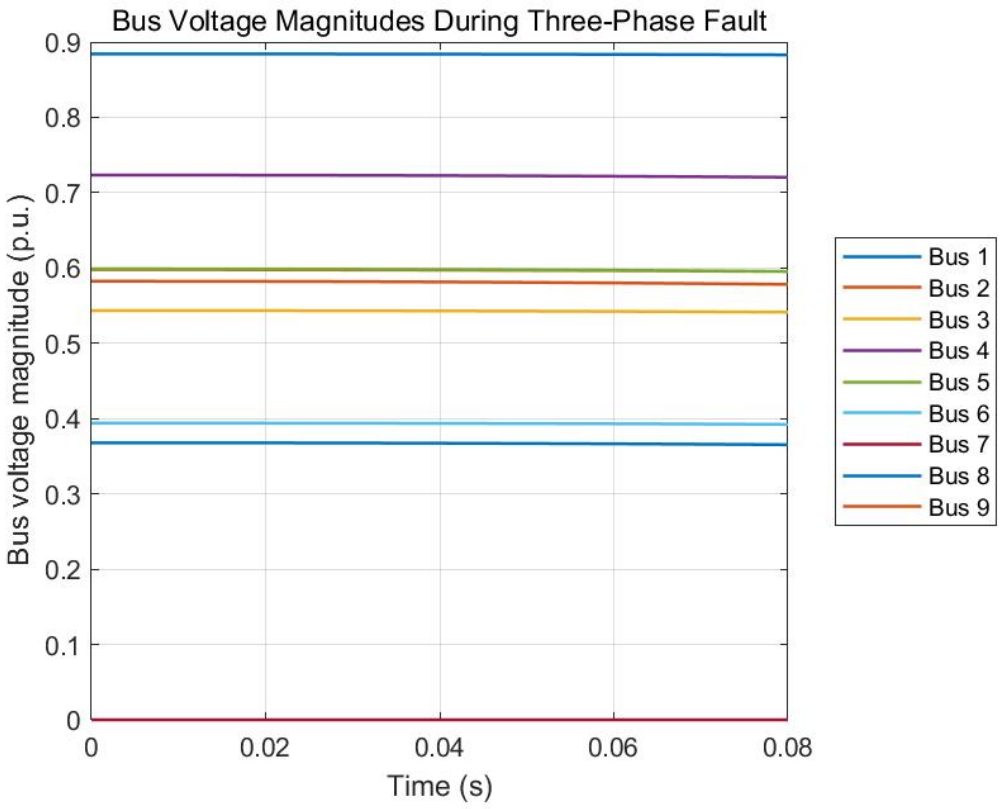


Figure 13. Bus voltage magnitudes during the three-phase fault.

8.3 Discussion

The simulation results show that the three-phase fault creates a severe disturbance to the power system. During the fault, the network voltage level is significantly reduced and the electrical power transfer capability is weakened. Since the mechanical input power cannot change instantly, the generators experience accelerating power and their rotor angles and rotor speeds increase.

The fault-on response also explains why fast fault clearing is important for transient stability. If the fault remains for too long, the generators may gain too much kinetic energy, and the relative rotor angles may grow beyond the synchronism limit after fault clearing.

Therefore, Task 5 shows that the three-phase fault causes voltage collapse at the faulted bus and accelerates the generator rotors during the fault-on period.

9. Fault Clearing and Post-Fault Stability Analysis (Task 6)

This section studies the post-fault dynamic response after the three-phase fault is cleared. The fault is cleared after 4 cycles in the 50 Hz system, corresponding to

$$t_{clear} = \frac{4}{50} = 0.08\ s$$

After fault clearing, the fault is removed and the line connected to Bus 7 in the implemented network topology is opened to form the post-fault network. Two damping cases are compared: $D = 0$ and $D = 10$. The purpose is to evaluate how damping affects rotor-speed oscillations, relative rotor-angle oscillations and post-fault stability.

9.1 Post-Fault Response without Damping (D = 0)

For the undamped case, the damping coefficient is set to zero:

$$D = 0$$

Therefore, the rotor-speed equation becomes

$$T_{J,i}\frac{d\omega_i}{dt} = P_{m,i} - P_{e,i}$$

Relative rotor angles are used to assess synchronism because they remove the common angular drift of all generators. Generator 1 is selected as the reference generator. The relative rotor angles are defined as

$$\Delta\delta_{21} = \delta_2 - \delta_1$$

$$\Delta\delta_{31} = \delta_3 - \delta_1$$

Figure 14 shows the rotor-speed responses after fault clearing for $D = 0$. The rotor speeds oscillate after the fault is cleared, indicating an exchange of kinetic and electrical energy between the generators and the network. Since no damping torque is included, the oscillations are poorly damped during the simulated time window.

Figure 15 shows the relative rotor angles using Generator 1 as the reference. The relative rotor angles remain bounded and do not exceed the synchronism limit during the 10 s simulation. Therefore, based on the relative rotor-angle criterion, the machines remain in synchronism for the 4-cycle clearing case. However, the response is poorly damped and is not desirable for practical operation.

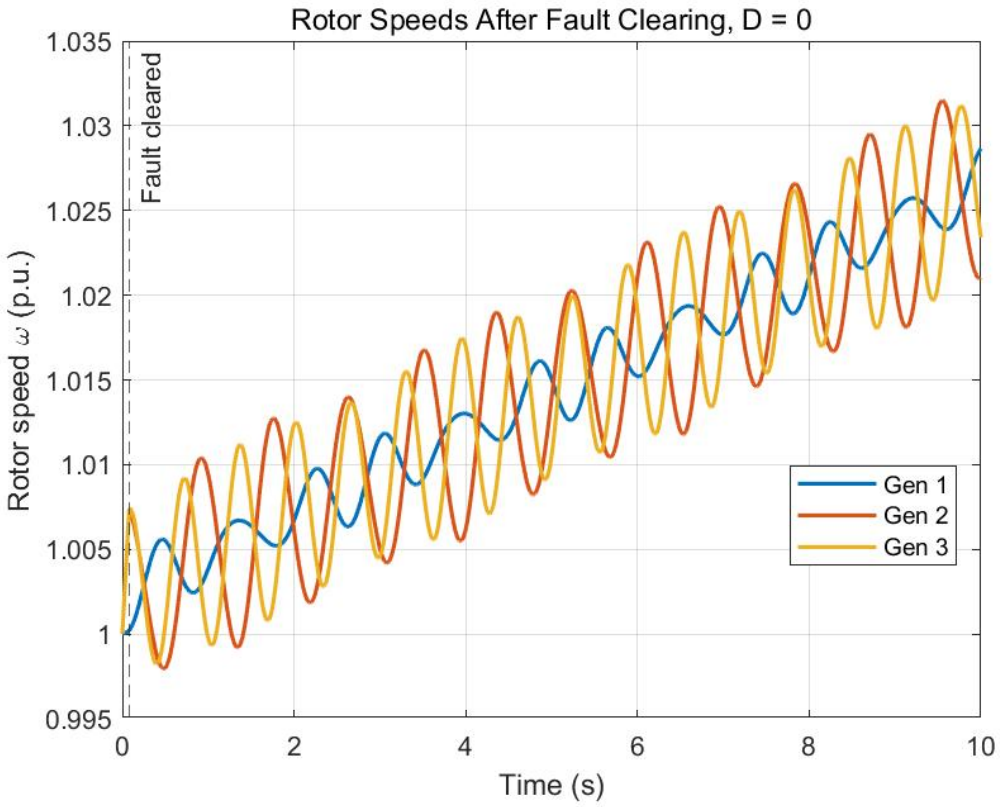


Figure 14. Rotor speeds after fault clearing, D=0.

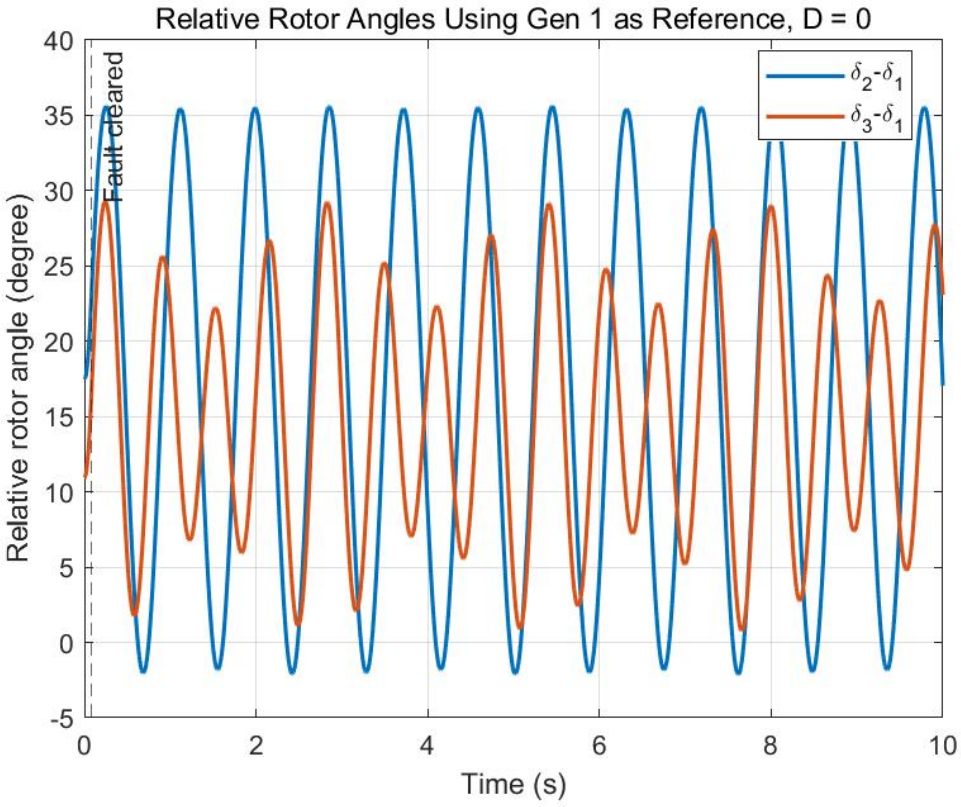


Figure 15. Relative rotor angles using Gen 1 as reference, D=0.

9.2 Post-Fault Response with Damping (D = 10)

For the damped case, the damping coefficient is set to D=10

The rotor-speed equation is

$$T_{J,i}\frac{d\omega_i}{dt} = P_{m,i} - P_{e,i} - D_i(\omega_i - 1)$$

The damping term is proportional to the rotor-speed deviation. When the rotor speed deviates from synchronous speed, this term provides an opposing torque and reduces the electromechanical oscillations.

Figure 16 shows the rotor-speed responses for $D = 10$. Compared with the $D = 0$ case, the rotor-speed oscillations are significantly reduced and approach a nearly steady value after the transient.

Figure 17 shows the relative rotor angles for $D = 10$. The relative rotor-angle oscillations are clearly damped and gradually settle to bounded values. This indicates that the damping term improves the post-fault dynamic response by reducing the oscillation amplitude.

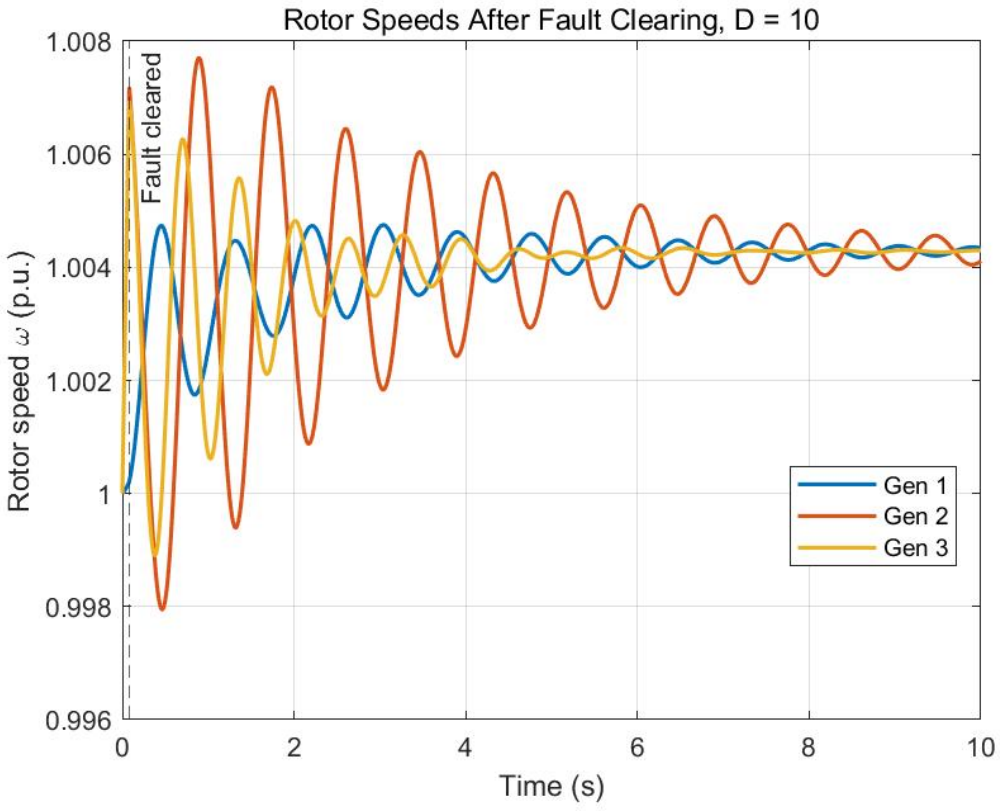


Figure 16. Rotor speeds after fault clearing, D=10.

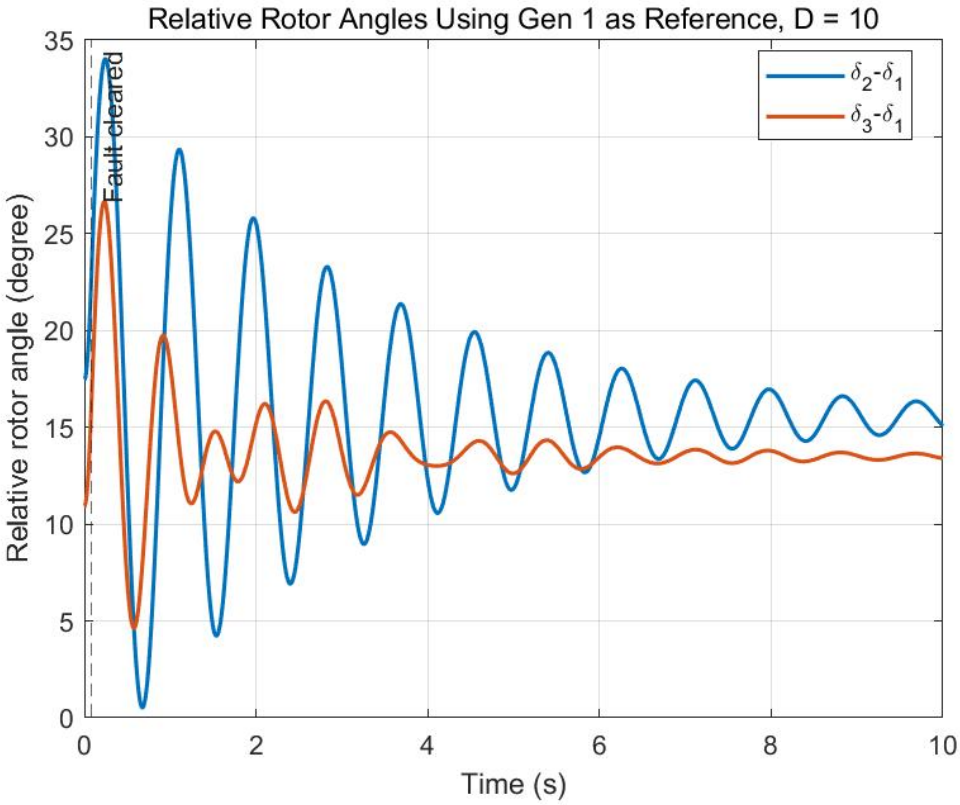


Figure 17. Relative rotor angles using Gen 1 as reference, D=10.

9.3 Comparison and Discussion

The comparison between $D = 0$ and $D = 10$ shows the effect of damping on the post-fault electromechanical response. In the undamped case, the rotor-speed and relative rotor-angle responses remain oscillatory and poorly damped. In the damped case, the oscillation amplitudes are reduced, and the system response becomes smoother.

The 4-cycle clearing time is much shorter than the critical clearing time calculated in Section 10. Therefore, the system remains in synchronism for both damping cases based on the relative rotor-angle criterion. However, the $D = 10$ case provides a better post-fault dynamic response because the oscillations decay more effectively.

Therefore, Task 6 demonstrates that damping improves post-fault dynamic performance by reducing rotor-speed and relative rotor-angle oscillations.

The post-fault bus voltage responses were also checked. After fault clearing, the bus voltages recover from the fault-on depression and oscillate around the post-fault operating level. Compared with the D=0 case, the D=10 case shows more strongly damped voltage oscillations.

10. Critical Clearing Time Estimation (Task 7)

This section estimates the critical clearing time of the system for the three-phase fault scenario under the undamped condition. The critical clearing time is the maximum fault clearing time for which the system can still maintain synchronism after the fault is cleared.

In this task, the damping coefficient is set to

$$D = 0$$

The relative rotor angles with respect to Generator 1 are used as the stability indicators.

10.1 Methodology

The stability assessment is based on the relative rotor angles of Generators 2 and 3 with respect to Generator 1. The relative rotor angles are defined as

$$\Delta\delta_{21} = \delta_2 - \delta_1$$

$$\Delta\delta_{31} = \delta_3 - \delta_1$$

During the CCT search, the system is considered unstable if either relative rotor angle exceeds 180 degrees during the simulation window:

$$|\Delta\delta_{i1}| > 180°$$

where $i = 2,3$. This criterion is used as a practical synchronism check for the numerical simulation.

The CCT is obtained by testing different clearing times. A coarse search is first used to identify the boundary between stable and unstable clearing times. Then, a bisection search is applied to refine the estimate of the critical clearing time.

10.2 CCT Calculation Results

The estimated critical clearing time under the undamped condition is

$$t_{CCT} \approx 0.31433\ s$$

Since the system frequency is 50 Hz, the corresponding number of cycles is

$$N_{CCT} = t_{CCT} f_0 = 0.31433 \times 50 = 15.72\ cycles$$

The fault clearing time used in Task 6 is 4 cycles:

$$t_{4-cycle} = \frac{4}{50} = 0.08\ s$$

Since

$$0.08\ s < 0.31433\ s$$

the 4-cycle clearing time is below the estimated CCT. This explains why the system remains in synchronism in the Task 6 simulations based on the relative rotor-angle criterion.

Figure 18 shows the stability result for different clearing times. The stable cases are below the estimated CCT, while the unstable cases occur when the clearing time is slightly above the stability boundary.

Figure 19 shows the relative rotor-angle responses near the estimated CCT. The stable case remains within the synchronism boundary, while the case just above the CCT reaches the instability criterion.

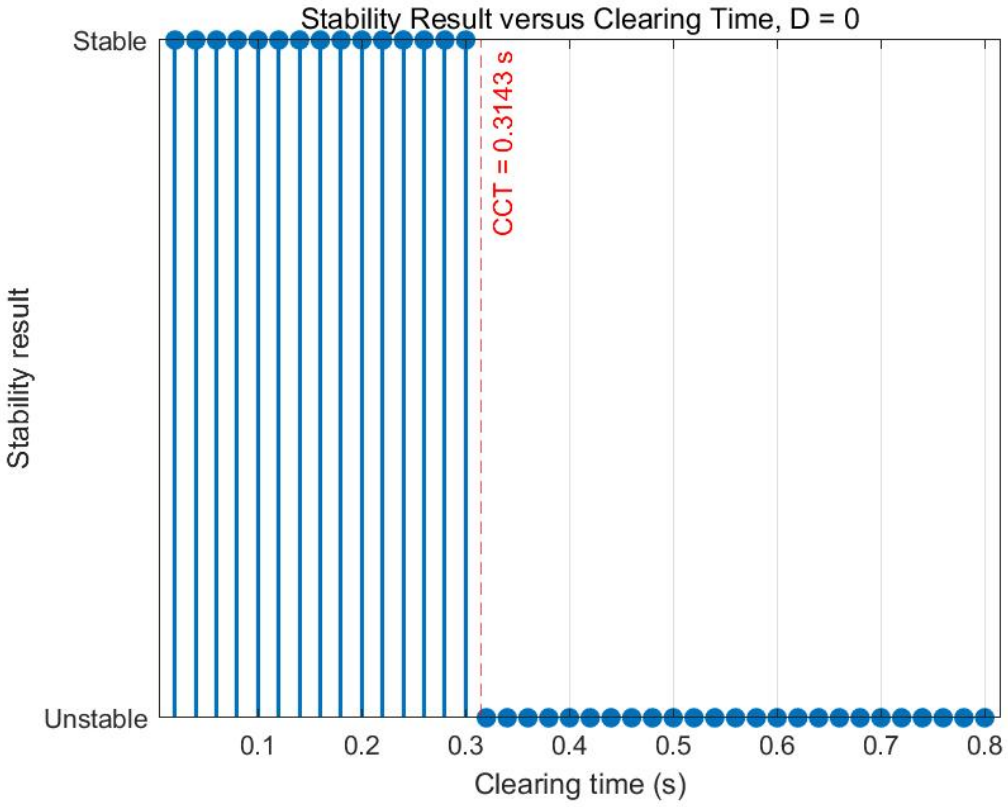


Figure 18. Stability result versus clearing time under D=0.

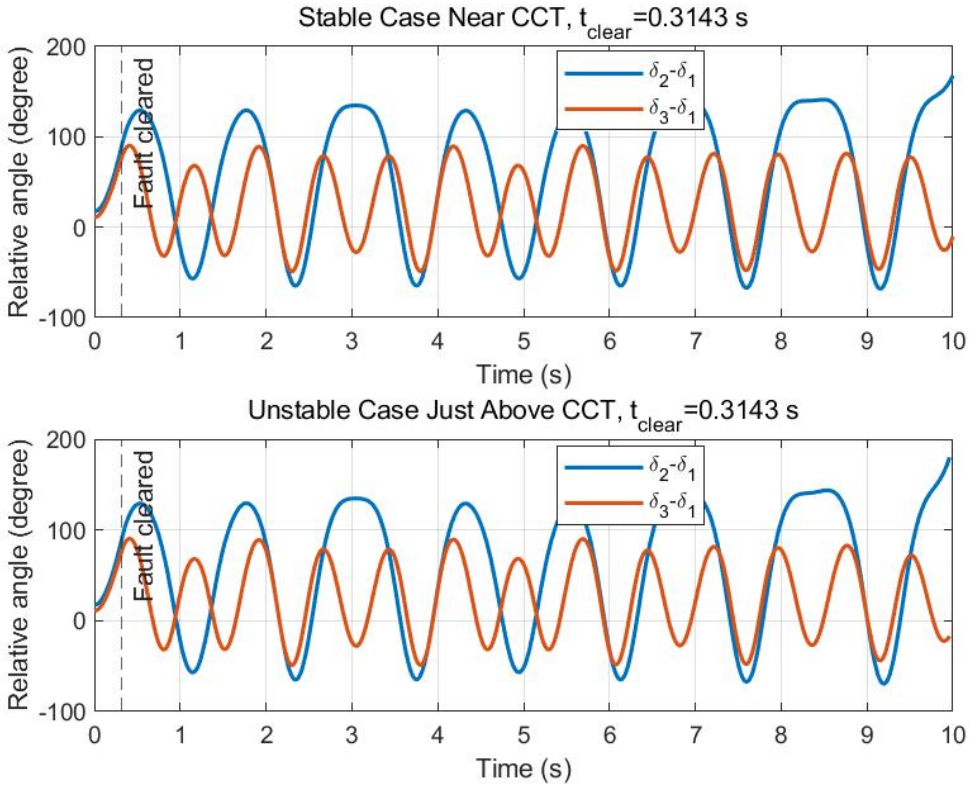


Figure 19. Relative rotor angles near the critical clearing time under D=0.

10.3 Discussion

The CCT result shows that fault clearing time has a strong influence on transient stability. If the fault is cleared before the estimated CCT, the generators can remain in synchronism after fault clearing. If the fault is cleared after the estimated CCT, the generators gain too much kinetic energy during the fault-on period, and the relative rotor angle may exceed the synchronism limit.

The estimated CCT in this simulation is approximately 0.31433 s, or 15.72 cycles at 50 Hz. The 4-cycle clearing time used in Task 6 is much shorter than this value, so the post-fault response remains stable according to the relative rotor-angle criterion.

It should be noted that the CCT value depends on the system operating condition, network topology, generator inertia, fault location, and damping model. Therefore, the calculated CCT should be interpreted as the stability limit for the specific fault scenario and modelling assumptions used in this project.

Therefore, Task 7 estimates the critical clearing time as approximately 0.31433 s, or 15.72 cycles at 50 Hz.

11. Debugging Experience and Implementation Notes

This section summarizes the main implementation issues encountered during the project and the corresponding solutions. These notes are included to explain how the simulation program was checked, corrected and organized during the project.

11.1 MATPOWER Path Configuration Issue

At the beginning of the project, MATLAB could not recognize MATPOWER functions such as runpf. This occurred because the MATPOWER folder had not been added to the MATLAB search path.

The problem was solved by adding the MATPOWER folder and its subfolders to the MATLAB path. After this, the standard case9 example was tested to confirm that the MATPOWER power flow functions could run correctly. This step was necessary before modifying the system data for the project tasks.

11.2 Automated Figure Output Issue

During the first simulations, the results were displayed only in MATLAB figure windows. This made it difficult to keep the figure numbering consistent in the report.

To solve this problem, the plotting sections of the code were modified so that each figure was saved automatically with a clear filename. The figure captions in the report were then matched with the saved figures. This helped organize the results for each task and reduced the risk of inserting the wrong figure into the report.

11.3 Dynamic Model Initialization Error

One important implementation issue was the conversion from the power flow solution to the dynamic model initial condition. The rotor angle cannot be taken directly as the terminal bus voltage angle. Instead, it must be calculated from the internal transient voltage.

The initialization procedure was checked using the following equations:

$$S_i = P_{G,i} + jQ_{G,i}$$

$$I_i = \left(\frac{S_i}{V_i}\right)^*$$

$$E'_i = V_i + jx'_{d,i}I_i$$

$$\delta_i = \angle E'_i$$

The inertia constant was also checked using the stored energy data:

$$H_i = \frac{W_{k,i}}{S_B}$$

After the initialization, the equilibrium mismatch was checked using

$$P_{m,i} - P_{e,i} = 0$$

The final results showed that the mismatch was zero for all three generators, confirming that the power flow solution was correctly converted into a valid dynamic equilibrium point.

11.4 Fault Modelling Implementation

Another implementation issue was how to represent different network conditions during the transient simulation. The system has different network configurations before the fault, during the fault, and after the fault is cleared. If the network data are modified repeatedly inside the differential equation function, the program becomes difficult to debug and may cause inconsistent results.

To avoid this problem, separate reduced admittance matrices were constructed for the fault-on and post-fault conditions before running the time-domain simulation. During the fault-on period, the three-phase fault was represented by adding a large shunt admittance at Bus 7:

$$Y_{fault} = 10^6\ p.u.$$

The fault-on simulation was first solved from $t = 0$ to the fault clearing time. Then, the final state at the clearing instant was used as the initial condition for the post-fault simulation. This two-stage simulation approach made the switching between fault-on and post-fault network conditions clear and easy to verify.

11.5 Network Topology Consistency Check

The project description specifies a three-phase fault near Bus 7 at the end of the line connected to Bus 7. In the implemented MATPOWER topology used for this report, the fault was represented as a solid three-phase fault at Bus 7 by using a large shunt admittance. After clearing, the fault was removed and the line connected to Bus 7 in the implemented topology was opened to form the post-fault network.

This approximation was used to keep the fault simulation consistent with the available network model. The same assumption was applied consistently in the fault-on simulation, post-fault simulation and CCT search.

11.6 Additional Implementation Notes

The dynamic simulations were solved using MATLAB ode45. The simulation program was organized into separate sections for power flow, dynamic initialization, fault-on simulation, post-fault simulation and CCT search. This modular structure made it easier to check the output of each task separately.

Consistent variable names were used in the code to reduce confusion. For example, generator internal voltage, rotor angle, mechanical power and electrical power were stored separately. This helped verify the dynamic equations and made the results easier to interpret in the report.

12. Conclusion

This project investigated the steady-state and transient dynamic behaviour of the IEEE 3-machine 9-bus power system using MATLAB and MATPOWER. The study covered steady-state power flow analysis, voltage sensitivity analysis, reactive power compensation, dynamic model initialization, three-phase fault simulation, post-fault stability analysis and critical clearing time estimation.

The base-case power flow calculation converged successfully and showed that the bus voltage magnitudes remain within the normal operating range under the base-case condition. The generator voltage set-point study showed that increasing generator voltage references raises the network voltage profile, while decreasing the voltage references lowers the bus voltage magnitudes. The reference bus angle study showed that changing the reference angle only shifts the numerical voltage angle reference and does not change the physical power flow condition.

The loading analysis showed that heavier system loading weakens the voltage profile. As the active and reactive loads increase, larger currents flow through the network impedances, resulting in larger voltage drops. Bus 9 was identified as the weakest voltage bus in the implemented system. The reactive compensation study showed that capacitive compensation at Bus 9 can improve the low-voltage condition, while excessive capacitive compensation may cause local overvoltage. Therefore, reactive compensation must be properly sized.

The dynamic model was initialized from the base-case power flow result. The generator currents, internal transient voltages, rotor angles and inertia constants were calculated consistently on the 100 MVA system base. The equilibrium check showed that $P_m - P_e = 0$ for all three generators, confirming that the base-case power flow solution was correctly converted into a valid equilibrium point for the dynamic model.

During the three-phase fault simulation, the voltage at the faulted bus collapsed close to zero and the generator rotors accelerated due to the reduction in electrical power transfer capability. After fault clearing at 4 cycles, the relative rotor angles remained bounded according to the relative rotor-angle criterion. The comparison

between $D = 0$ and$D = 10$ showed that damping significantly improves the post-fault dynamic response by reducing rotor-speed and relative rotor-angle oscillations.

The critical clearing time under the undamped condition was estimated as

$$t_{CCT} \approx 0.31433\ s$$

which corresponds to

$$N_{CCT} = 15.72\ cycles$$

at 50 Hz. Since the 4-cycle clearing time used in the post-fault simulation is

$$t_{4-cycle} = 0.08\ s$$

and this value is below the estimated CCT, the system remains in synchronism for the tested 4-cycle clearing case under the modelling assumptions used in this project.

Overall, the results show how voltage control, loading level, reactive compensation, damping and fault clearing time influence the steady-state and transient stability performance of a multi-machine power system.